TERASMART

PROJECT DOCUMENTATION

NISR Hackathon Competition 2026
Agricultural Productivity Track

Team RwandaRise

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1. Executive Summary

TeraSmart is a Streamlit-based agricultural decision-support prototype created by Team RwandaRise for the NISR Hackathon 2026. It supports two roles: Farmer and Planner. The system combines field registration, historical-survey-based crop-production estimates, task planning, crop observations, basic financial recording, district weather forecasts and optional AI assistance. Planner dashboards summarize records entered into the application and offer downloadable planning outputs.

TeraSmart is a distinct project from IhingaAI. Its packaged prediction engine uses frozen model artifacts originally developed for IhingaAI, but the TeraSmart application, interface, database and intended workflows are separate. Predictions are experimental and must not be interpreted as official NISR forecasts or guaranteed harvests.

2. Problem, Objectives and Users

2.1 Problem statement

Farmers may manage crop information, expenditure, work schedules and weather decisions across disconnected records. Without consolidated tools, it is difficult to monitor a field consistently or revisit previous production estimates. Agricultural planners also need clear local operational summaries without incorrectly presenting small application datasets as representative district or national statistics.

2.2 Project objectives

- Provide crop-production estimates for supported crop and season combinations.
- Enable field registration, updating and activity monitoring.
- Record selected farm income and expenses, including a simple cash-flow balance.
- Show district-reference weather conditions and rainfall probabilities.
- Support planners with aggregated registered-field insights and exportable reports.
- Offer optional, clearly qualified AI-assisted answers.

2.3 User roles

Role	Main responsibilities
Farmer	Maintain fields, request harvest estimates, record tasks, observations and finances.
Planner	Review aggregated application data, production forecasts and planning reports.

3. Data Sources and Governance

Source	Use	Qualification
NISR SAS 2025-derived model data	Historical patterns underlying frozen prediction artifacts	Not live field observations or official forecasts
SQLite database	Demo users, fields, forecasts, observations, tasks and ledger	Application-specific and not nationally representative
Open-Meteo	Weather forecasts from approximate district coordinates	Not a plot-level weather station
Google Gemini (optional)	General and context-aware text assistance	Generated content requires checking

3.1 Reference and prediction files

The data/ folder contains crop_catalog.json, district_labels.json and district_weather_points.json alongside two serialized prediction artifacts: IhingaAI_Frozen_Model_v1.pkl and IhingaAI_Frozen_Extension_v1.pkl. The embedded file names reflect the model provenance, not the TeraSmart project name. The model-development datasets are not included in the supplied TeraSmart ZIP.

3.2 Data management

User records are stored locally in SQLite. Access is role-based and password hashes are used for login. A local database should be treated as demonstration storage, not a production-ready multi-user platform. Real credentials, private API keys and identifiable farmer records must be excluded from any public repository. Survey data redistribution should follow the provider's permissions.

4. Predictive Modeling Methodology

4.1 Estimation flow

The Farmer chooses a registered field and requests a crop-production estimate. TeraSmart sends the crop code, agricultural season, district, farmer category and plot area to core/router.py. The router validates the request and dispatches it to one of the two frozen model pipelines. The returned estimated production and lower/upper range are displayed and may be saved for later review.

4.2 Supported scope and checks

The application catalog supports ten crops through the combined model router. Availability is determined by crop-season entries in crop_catalog.json; not every crop is supported in every season. Prediction scope is limited to small-scale farmer categories. Invalid locations, non-positive areas, unsupported crop-season combinations and certain out-of-range areas are rejected rather than silently assigned a forecast. Some results carry warnings for unusual areas or data-sparse combinations.

4.3 Model interpretation

The model artifacts reuse historical patterns derived from the NISR 2025 Seasonal Agricultural Survey. The response may contain an indicative interval around predicted production in kilograms; these bounds are not a guarantee for a particular field. Saved estimates remain historical records: changing a field does not automatically recalculate a previously saved prediction. Users must request a new forecast after material field changes.

4.4 Evaluation and limitations

The packaged application includes trained model files and inference logic but not full training notebooks or evaluation tables. Consequently, this report does not attribute a new accuracy score to TeraSmart or claim independent model validation. Before agricultural deployment, predictions should be evaluated per crop, season, farm scale and geographic setting using independent observations.

4.5 High-level architecture

Layer	Implementation
Presentation	Streamlit interface, role-specific navigation and Plotly charts
Application	app.py and core/ Python modules
Prediction	Serialized scikit-learn-compatible frozen pipelines
Persistence	SQLite local database and static JSON reference files

External services	Open-Meteo forecasts and optional Gemini text generation

5. Farmer Features and Workflow

5.1 Fields and crop details

Farmers can register fields and edit name, crop, season, district, area, farmer category, planting date, seed variety and fertilizer selection. Seed choices are Local, Improved and Not sure. Fertilizer choices are Organic (manure/compost), Inorganic (mineral/NPK) and None / not applied. Existing databases are migrated to support added fields.

5.2 Forecasting and monitoring

The harvest estimator generates supported model-based production ranges. Saved forecast histories help farmers revisit earlier estimates. A monitoring page stores crop stage, condition and observations; it supports practical tracking but does not by itself diagnose plant disease.

5.3 Activities and financial records

Users can create and complete farm activities. The dashboard highlights overdue tasks in red and upcoming or current tasks in green. Financial entries record costs and sales in Rwandan francs. The displayed farm balance is calculated as recorded sales minus recorded expenses, not full economic profit or an audited farm account.

5.4 Weather and AI assistant

District weather uses Open-Meteo predictions based on reference points. Daily values include rain amounts and maximum precipitation probability, which are forecasts rather than recorded rainfall. Gemini can answer general or farming questions when an API key and internet access are available. Without Gemini, the assistant offers limited rule-based farm guidance and may be unable to answer general questions.

5.5 Typical farmer journey

- Sign in to the Farmer workspace and review the dashboard.
- Register or edit a field, including seed and fertilizer information.
- Generate and save a supported production estimate.
- Record observations, activities and cash transactions.
- Review weather context and optional AI support before planning work.

6. Planner Features and Reporting

6.1 Planner dashboard

The Planner workspace aggregates information from registered TeraSmart fields, rather than the complete agricultural population. It provides overview metrics, production forecasts and crop-season views to help interpret the demonstration data.

6.2 Forecast aggregation

For planner totals, the application selects the latest saved estimate per field, avoiding double counting when a farmer has forecast the same field more than once. Fields with no saved estimates do not contribute to the forecasted production total. Available filters support comparing registered data by relevant groups.

6.3 Reports and source transparency

Planner reporting offers downloadable Markdown and CSV outputs. The Data Sources page explains historical model inputs and weather and AI dependencies. These reports are operational summaries of participation in TeraSmart, not official district-level yields, NISR publications or representative sample estimates.

7. Implementation and Repository

Component	Purpose
app.py	Streamlit entry point, authentication, user flows and dashboards
core/predictor.py	Baseline frozen-model inference and input validation
core/router.py	Crop selection, scope checks and extension-model routing
data/	Reference JSON files and two model artifacts
storage/terasmart.sqlite3	Supplied local demo records
assets/	Page imagery and theme styling
requirements.txt	Pinned or bounded Python dependencies
.streamlit/config.toml	Streamlit presentation configuration

The GitHub repository is https://github.com/fraterneniyo/TeraSmart. The submission approach is repository-only: judges should download or clone the files and run the application locally. The model .pkl files together exceed 100 MB, so Git LFS is appropriate for GitHub hosting.

8. Installation and Demonstration

8.1 Local setup

Use Python 3.11 or 3.12. Install Git LFS before cloning to retrieve the large model files. Then, in a terminal inside the project folder, run the following commands:

```
git lfs install
git clone https://github.com/fraterneniyo/TeraSmart.git
cd TeraSmart
git lfs pull
python -m pip install -r requirements.txt
python -m streamlit run app.py
```

Open the local URL shown by Streamlit, usually http://localhost:8501. The repository must preserve the data/ folder, model files and application folder structure. When the sample database is included in storage/, it supplies existing demonstration records.

8.2 Demonstration credentials

Role	Telephone	Demo password
Farmer	0790888558	Frat@2026
Planner	079088559	Just@2026

These credentials were supplied by the team for demonstration. They have not been authenticated in a running browser during this documentation review. Use only synthetic accounts and change credentials before any real-world deployment.

8.3 Optional configuration

Google Gemini questions require GEMINI_API_KEY, provided securely via a local environment variable or a private Streamlit secrets file. Planner registration can require PLANNER_SETUP_CODE.

8.4 Suggested evaluation walk-through

Farmer: log in, review records, edit a field, generate a forecast, inspect overdue activities, enter a transaction and check rainfall probability. Planner: log in, view registered-field summaries, review latest forecast totals and download a report.

9. Testing, Limitations and Future Work

9.1 Inspection and verification

This documentation is based on code and data inspection of the supplied TeraSmart ZIP. The implementation contains functions for both roles, database setup and migration, task status indicators, finance summaries, weather requests, crop routing, AI fallback and planner reporting. This is not equivalent to completing end-to-end browser testing, verifying every demo login or testing every external service.

9.2 Known limitations

- Harvest estimates are experimental and constrained by model eligibility checks.
- The app includes historical-survey-trained models but no independently verified current-season forecast accuracy.
- Small demonstration records cannot justify district-wide or national inference.
- Weather is district-reference data; conditions can differ at individual fields.
- AI responses can be incorrect; general questions require a configured Gemini service.
- SQLite and local files lack the durability and controls needed for a production system.
- Manual evaluation of session security, user permissions and exports remains necessary.

9.3 Recommended improvements

Key next steps are persistent managed storage, comprehensive user-access testing, crop-specific model validation, usability trials with farmers, agronomist review of advice, and more reliable handling of external-service outages.

10. Conclusion

TeraSmart demonstrates how historical agricultural data and lightweight digital workflows can support practical field-level decisions and operational planning. Its distinguishing contribution is the integration of harvest estimation, editable field records, task monitoring, basic cash-flow tracking, weather context and accessible planner views within a two-role prototype. Responsible interpretation of model results and stronger security and validation are essential before real-world expansion.

11. References

National Institute of Statistics of Rwanda (NISR). Seasonal Agricultural Survey 2025. https://www.statistics.gov.rw/

Open-Meteo. Forecast API and documentation. https://open-meteo.com/

Google AI for Developers. Gemini API. https://ai.google.dev/

RwandaRise. TeraSmart source repository. https://github.com/fraterneniyo/TeraSmart